

01

Product Requirements & Scope

What TrustHome PH must do, for whom, and what it deliberately does not do. This is the product source of truth. Every feature, test case and panel answer traces back to it.

Revision	Rev 2.0 — 1 October 2026
Supersedes	Build Instructions (Listing, Verification & Booking Modules); the "Proposed FRs" sheet of the workbook
Stack baseline	React + Firebase Spark (Auth, Firestore, Hosting) + Cloudinary Free + Vercel Hobby API + GitHub Actions. No paid plan, no credit card.
Audience	Whole team, AI coding assistants, panel reviewers
Team	Andresa (Scrum Master) · Carillo · Laurente · Prades · Solpico

READ THIS FIRST

TrustHome PH is a verified rental-and-sale marketplace for the Philippines, with a secondary Metrobank bank-acquired property catalog. Since the first Build Instructions, the scope has changed in five ways. Listings now declare a purpose (short-term rent, long-term rent, or sale). Exact addresses and verification documents are private. Booking confirmation must run on a trusted server. Saved Searches is retired. And the whole system must run without any paid plan or credit card. Requirements FR-001 to FR-029 and the NFRs below replace every earlier list. Each one shows its implementation status as of 1 Oct 2026.

CONTENTS

- | | | | |
|---|----------------------------|--------------------------------|--|
| 1. Purpose and how to use this document | 4. Core user journeys | 7. Booking state machine | 10. Honest limitations (use these exact words) |
| 2. Product summary | 5. Functional requirements | 8. Trust & Fairness Score | 11. Glossary |
| 3. Actors and roles | 6. Listing intake rules | 9. Non-functional requirements | |

Document set: **01 Product Requirements & Scope** · 02 System Architecture & Data Design · 03 Infrastructure, Setup & Deployment · 04 Security & Privacy · 05 UI/UX Design System · 06 Roadmap & Release Plan · 07 QA & Testing Strategy

1. Purpose and how to use this document

This document answers three questions: **what must the system do**, **who is it for**, and **where does scope stop**. It is not a technical design. That lives in *02 System Architecture & Data Design*.

Source-of-truth rule. A feature can be built or merged only if:

1. it maps to a requirement ID in Section 5 (or a new one is added here first);
2. it serves a named actor in Section 3;
3. it has at least one test case in the workbook ([TrustHome_PH_Use_Cases_and_Test_Cases_v2.xlsx](#));
4. any limitation it has is written into Section 10, *Honest limitations*.

BEST PRACTICE · Requirements describe **observable behaviour** ("a sale listing cannot be booked"), never implementation ("add an `if` in `BookingPage.jsx`"). That way a requirement survives a stack change. If the backend later moves to Laravel, this document should not need edits.

1.1 What changed since the original Build Instructions

Area	Before	Now (Rev 2)	Why
Listing type	Monthly rent only	Purpose: For rent (short-term nightly / long-term monthly) or For sale	Owners asked for it; sale listings appear in the market
Description	150–400 words required	Non-empty, max 400 words ; 150 is a recommendation	Panel feedback: concise factual listings were blocked
Exact address	"Can be hidden"	Always private (listingPrivate); public map shows a ~5 km area	Safety and privacy; handed to renter only after Confirmed
Verification documents	Unsigned Cloudinary URL on the listing	Private assets ; short-lived links for owner/admin only	Finding F-16: anyone holding the URL could open an ID
Booking confirmation	Client-side transaction	Trusted server endpoint with lock + transaction	Finding F-05: overlaps possible in deployed app
Completion	Owner or renter	Renter only + automatic fallback after end date	Duplicate buttons confused users
Saved Searches / paid alerts	Planned paid feature	Retired ; URL redirects to Browse; data kept	Out of scope for zero-billing MVP
Notifications	Booking requests broadcast to admins	Admins get disputes and new-listing submissions only	Booking activity is private to renter and owner
Cost model	"Free tier"	Zero-billing constraint (NFR-COST-01): nothing that needs a credit card	No team member has a card accepted by Google Blaze

2. Product summary

Problem. Rental listings in the Philippines are full of fake owners, recycled photos, and deposit scams. Renters cannot tell who really controls a property. Honest owners cannot prove they are honest.

Solution. TrustHome PH combines four things:

- **Document-reviewed listings.** A person checks ownership plausibility before a listing goes public.

- **Conflict-free booking.** The server enforces it, so two renters can never hold the same dates.
- **An accountability trail.** Deposit references, disputes, and permanent public flags for founded cases.
- **A Trust & Fairness Score.** Listings are ranked by completed bookings, ratings and price fairness, not by recency or advertising.

A secondary module shows a **bank-acquired (ROPA) property catalog**, scraped from Metrobank's public pages, with a loan calculator.

2.1 In scope (MVP, current release)

- Email/password and Google sign-in; one account type for renters and owners; separate admin role.
- Listing intake with purpose, photos, two verification documents, showing windows, optional approximate map pin.
- Admin review: approve, request changes, reject. Each decision carries a note.
- Verified-only public search, sorted by Trust & Fairness Score.
- Booking requests, owner confirmation (trusted API), decline, deposit reference, renter completion, automatic completion fallback.
- Ratings, disputes, public accountability flag, trust score and fairness label.
- In-app notifications, My Activity hub, Settings (appearance and accessibility).
- Metrobank catalog, admin scraper-job management, loan/amortization calculator.
- Legal and information pages (Privacy Policy, Terms of Use, Data & Compliance, IP Infringement Reports).

2.2 Out of scope (state this plainly)

Item	Status	Reason
Holding or escrowing deposits	Never in MVP	Needs a BSP-regulated payment partner
Legal title search (Registry of Deeds / LRA)	Never in MVP	Review is plausibility only
Email, SMS, push notifications	Future	Free tiers are limited and most need billing; in-app only for now
BDO and Landbank scrapers	Future	Selectors and API shape not verified; keep disabled
Firebase Storage	Future, only if billing is ever approved	Requires Blaze since Sept 2024
Native mobile app	Future	Responsive web first
Saved Searches and paid alerts	Retired	Removed in the 30 Sept cleanup

3. Actors and roles

Actor	How they get the role	Can do	Cannot do
Guest	Not signed in	Browse verified listings, view details (approximate map only), bank catalog, loan calculator, legal pages	Book, list, rate, see exact addresses or documents

Actor	How they get the role	Can do	Cannot do
User (renter and owner)	Registers; <code>role: "user"</code> , <code>status: "active"</code>	Everything a guest can, plus: create listings, request bookings, confirm/decline requests on own listings, log deposit references, complete, rate, dispute	Approve listings, read others' private data, change own role or status
Suspended user	Admin action or founded dispute	Sign in and see a suspended-account screen	Any write action; their listings are hidden from search
Admin	Set manually in Firestore by the team; <code>role: "admin"</code> , <code>status: "active"</code>	Review listings and documents, resolve disputes, suspend/reinstate users, manage scraper jobs, see admin alerts	Confirm bookings for owners, edit ratings, create notification broadcasts
System jobs	GitHub Actions (Python, service account) and the trusted API	Scrape catalog, complete expired bookings, recompute trust scores, confirm bookings, sign document links	Act without an audit trail

BEST PRACTICE · Least privilege. Admin is a role for reviewing, not a super-user. Admins cannot impersonate owners or fake bookings. This keeps the trust story defensible: *"even our admins cannot manufacture a completed booking."*

4. Core user journeys

4.1 Owner: list a property and get verified

1. Register and sign in.
2. **Create Listing** (multi-step): purpose and price period → details → photos (min 4) → documents (ownership + government ID) → showing windows → optional map pin → review and submit.
3. The listing becomes `pending`. It is not public. An admin alert ("new listing submitted") is written in the same batch.
4. An admin reviews it. The admin compares the name on the document with the name on the government ID and the account, then checks document plausibility.
5. Outcome: **Verified – document reviewed** (public), **Changes requested** (the owner edits and resubmits), or **Rejected** (final). The owner is notified with the review note.
6. The owner manages incoming requests in **My Activity → Booking Requests**: Confirm (through the trusted API) or Decline.
7. The deposit is paid **directly** by the renter (GCash or bank transfer). TrustHome never holds money.

4.2 Renter: find, book, and stay protected

1. Browse verified listings (default sort: Trust & Fairness Score). Filter by city, price, type and purpose.
2. Open a listing. See the trust score with a plain-language cue, the fairness label, photos, an approximate map, and reviews.
3. Request dates. The client pre-checks overlaps for fast feedback. The server re-checks at confirmation.
4. When **Confirmed**: the exact address is revealed, and the renter logs the deposit transaction reference.
5. After the stay or viewing: the renter marks **Completed** (or the daily job completes it after the end date). The renter rates the stay.

6. If the owner does not honour a Confirmed booking, the renter raises a **Dispute**. Founded disputes suspend the owner and leave a permanent public flag.

4.3 Admin: keep the marketplace honest

Listing review queue (with private document preview) → dispute queue (Founded / Dismissed with notes) → user management (suspend/reinstate, with confirmation) → bank catalog status and scraper jobs → admin notifications (disputes and new submissions only).

5. Functional requirements

Status key: Done implemented and tested in dev · Partial implemented, live verification or deployment pending · Open blocking defect or not built · Retired

ID	Requirement (the system shall...)	Priority	Status
FR-001	Let users register and sign in with email/password or Google, keep one profile per UID, and block suspended accounts from protected actions.	High	Done
FR-002	Require purpose, property type, city/area, private exact address, bedrooms, bathrooms, size (sqm), price, availability date, amenities, a non-empty description of at most 400 words, and at least 4 photos. 150 words is shown as a recommendation.	High	Done
FR-003	Require one ownership document and one government photo ID per listing. Keep the listing pending until an admin approves, requests changes, or rejects it with a note.	High	Done
FR-004	Show only verified listings from active owners in public search, with filters (city, price range, type, purpose) and default sort by trust score.	High	Done
FR-005	Support the booking state machine in Section 7 exactly, with no other statuses.	High	Done
FR-006	Reject any confirmation whose date range overlaps an existing Confirmed booking on the same listing, including when two confirmations race.	Critical	Open F-05
FR-007	Let the renter log a deposit transaction reference on their own Confirmed booking.	Medium	Done
FR-008	Let only the renter manually mark a booking Completed. Auto-complete Confirmed bookings whose end date has passed. Prompt for a rating on completion.	Medium	Done
FR-009	Let the renter dispute a Confirmed or recently Completed booking. Submission sets the booking to Disputed and stores its prior status. Dismissed restores the prior status. Founded keeps Disputed, suspends the owner, and adds a public flag. One open dispute per booking.	High	Done
FR-010	Compute a Trust & Fairness Score per listing from completed bookings, ratings, and price fairness (Section 8). Show it with a plain-language cue.	High	Done
FR-011	Flag for manual review when the same renter completes 3 or more bookings with the same owner within 30 days.	Medium	Done
FR-012	Give owners a dashboard of listings, pending requests, confirmed bookings and completed stays.	Low	Done
FR-013	Deliver in-app notifications to the right recipient only: owner for booking requests, renter for decisions, reporter for dispute outcomes.	Medium	Partial
FR-014	Scrape the Metrobank public ROPA catalog on a schedule, upsert by bank + reference number, mark missing items delisted, and show "Last verified" dates.	Medium	Done

ID	Requirement (the system shall...)	Priority	Status
FR-015	Provide a loan/amortization calculator pre-filled from a bank property's price. Handle a 0% rate.	Medium	Done
FR-016	Give admins a dashboard, user management (suspend/reinstate with confirmation), and scraper-job management.	Medium	Done
FR-017	Saved searches and alerts. <code>/saved-searches</code> redirects to Browse; stored records are kept.	—	Retired
FR-018	Show public profiles with trust signals and a founded-dispute flag.	Low	Done
FR-019	Render admin routes only for profiles with <code>role: admin</code> and <code>status: active</code> . Missing or unreadable profiles fail closed with a retry.	High	Partial
FR-020	Limit the admin feed to disputes and new-listing submissions. Isolate user feeds by rules.	High	Partial
FR-021	Provide one My Activity page (Listings, Bookings, Requests tabs) and Settings with System/Light/Dark plus persisted Larger text, High contrast, and Reduce motion.	Low	Partial
FR-022	Store an optional map pin publicly only as a 5-character geohash (~5 km). Keep the exact address in <code>listingPrivate</code> , readable by owner and admins only.	Medium	Partial
FR-023	Show per-listing photo, city, type, bedrooms, price, status, review note and pending-request count in My Listings. A failed count load must not display as zero.	Medium	Partial
FR-024	Reveal the exact address to the renter only after the booking is Confirmed.	High	Done
FR-025	Require a listing purpose: For rent (short-term nightly or long-term monthly) or For sale (one-time asking price). Sale listings cannot be booked. Fairness comparisons use only rentals with the same price period.	High	Partial
FR-026	Create an admin "listing submitted" alert in the same write batch as the pending listing, linking to the review item.	Medium	Partial
FR-027	Run booking confirmation only on the trusted API: verified ID token, owner and active-status check, Firestore transaction with a per-listing lock. Clients cannot write <code>Confirmed</code> .	Critical	Open v1.4
FR-028	Store verification documents as private Cloudinary assets. Deliver them only through short-lived signed links issued to the owner or an admin. Never put a document URL in public data.	Critical	Open F-16
FR-029	Make notification rows navigate to their link and mark read on open. Support Clear and Clear read.	Low	Done

NOTE · Status reflects the Current Progress Report of 1 Oct 2026. "Partial" usually means *built and unit-tested, but authenticated live testing has not been done*. That is a QA task, not a coding task (see 07 QA & Testing Strategy).

6. Listing intake rules

6.1 Fields

Field	Type	Rule	Notes
Purpose	Choice	Required: <code>rent</code> or <code>sale</code>	Drives price label and bookability
Price period	Choice	Required if rent: <code>nightly</code> or <code>monthly</code>	Sale uses a one-time asking price
Property type	Select	Room / Studio / Apartment / House / Condo	Used for fairness comparables
City and barangay/area	Text + select	Required	Public
Exact address	Text	Required; private	Stored in <code>listingPrivate/{id}</code>
Map pin	Map	Optional	Public value = 5-char geohash only
Bedrooms / bathrooms	Integer ≥ 0 / ≥ 1	Required	Studio = 0 bedrooms
Floor area (sqm)	Number > 0	Required	Fairness size band $\pm 20\%$
Price (PHP)	Number > 0	Required	Whole pesos; no decimals in UI
Availability date	Date	Required; today or later	—
Amenities	Multi-select	At least one	Parking, Wi-Fi, furnished, pets allowed...
Description	Text	Non-empty, ≤ 400 words	Recommend 150+ factual words
Photos	Images	≥ 4 (recommend 8–15)	JPG/PNG/WebP, ≤ 10 MB each
Showing windows	Day + start/end	≥ 1 valid window; end $>$ start	Shown to renters when requesting
Ownership document	Image/PDF	Required, private	See 6.2
Government photo ID	Image/PDF	Required, private	Name must match the ownership document

6.2 Accepted verification documents

Priority	Document	What it proves
Primary	Transfer Certificate of Title (TCT), Condominium Certificate of Title (CCT), or Deed of Sale	Strongest evidence of ownership
Primary	Recent Real Property Tax receipt or Tax Declaration	Person is the recorded taxpayer
Supporting	Contract to Sell / mortgage statement	Person is the buyer/borrower
Supporting	Lease authority (Special Power of Attorney from the owner)	Agent is allowed to lease

Priority	Document	What it proves
Always	Government photo ID: PhilID/ePhilID, passport, driver's licence, UMID, PRC ID	Identity; the name must match

WATCH OUT · Never ask for more than you need. **Do not** request TIN, SSS numbers, or bank statements, and **do not** OCR or store ID numbers in Firestore. The admin only needs to *see* the documents. This keeps TrustHome aligned with the proportionality principle of the Data Privacy Act (see *04 Security & Privacy*).

7. Booking state machine

Only these four statuses exist: **Pending**, **Confirmed**, **Completed**, **Disputed**. Declining a request deletes the Pending booking. There is no **Cancelled** status.

From → To	Who	Where it is enforced	Side effects
(new) → Pending	Renter	Client write; rules check listing is verified, rentable, not own listing, valid dates	Owner notified (best-effort)
Pending → Confirmed	Listing owner	Trusted API only (FR-027). Rules deny any client write of Confirmed	Overlap re-check in transaction; renter notified; address handoff created
Pending → (deleted)	Owner (decline) or renter (withdraw)	Rules	Renter notified on decline
Confirmed → Completed	Renter, or daily job after end date	Rules / GitHub Actions job	Rating prompt; counts toward trust score
Confirmed or Completed → Disputed	Renter (Completed: within the dispute window)	Rules	Prior status stored; admins alerted
Disputed → prior status	Admin (Dismissed)	Rules	Reporter notified
Disputed stays	Admin (Founded)	Rules + admin action	Owner suspended; public flag; reporter notified

Overlap rule. Use half-open ranges: a booking occupies `[startDate, endDate)`. Two ranges overlap when `existing.start < requested.end` **and** `requested.start < existing.end`. A check-out day can therefore be the next guest's check-in day. Only **Confirmed** (and Disputed-from-Confirmed) bookings block dates. Pending requests may compete.

WATCH OUT · **Decision needed (TC-DISP-009):** how long after Completed can a renter dispute? The proposed default is **7 days**. Record the team's decision here and in the rules.

8. Trust & Fairness Score

$$\text{trust} = 0.4 \times \text{bookingScore} + 0.4 \times \text{ratingScore} + 0.2 \times \text{fairnessScore} \quad (\text{each component } 0-1)$$

- **bookingScore**: completed bookings, recency-weighted (1.0 if ended within 6 months, else 0.5), then normalized against the highest listing so it stays 0–1.
- **ratingScore**: mean rating ÷ 5. With **no ratings**, use a neutral 0.5 and show "New listing". Never NaN.
- **fairnessScore**: compare price per sqm to the **median** of comparables (same city, same type, floor area within ±20%, same purpose and price period, verified, excluding the listing itself). Labels: **Above market** (> +15%), **At market** (±15%), **Below market** (< -15%). With fewer than 3 comparables, the label is **Insufficient data** and the neutral score is 0.5.
- **Anti-gaming**: only Completed bookings count. The same renter-owner pair with ≥ 3 completions in 30 days sets `manualReviewRequired` and freezes that listing's booking component until an admin clears it.

NOTE · The normalization of bookingScore and the neutral 0.5 defaults are Rev 2 clarifications. They stop the score from going above 1 or becoming NaN. Check them against the current scoring code (TC-TRUST-004, TC-FAIR-002). If the code differs, change the code or this text, but do not leave them disagreeing.

SAY THIS TO THE PANEL · "The score is computed by a scheduled job from Completed bookings only. A user cannot raise it by creating requests, and an owner cannot confirm their own fake bookings because confirmation runs on our server and checks ownership."

9. Non-functional requirements

ID	Category	Requirement	Measure
NFR-COST-01	Cost	No component may require a credit card or paid plan	Firebase Spark, Cloudinary Free, Vercel Hobby, GitHub Free
NFR-SEC-01	Security	Deny by default. Every collection has explicit rules; the last match is deny-all	Rules emulator suite passes
NFR-SEC-02	Security	Service-account keys are per consumer, stored only as secrets, and rotated after exposure	F-01 closed, history scan recorded
NFR-PRIV-01	Privacy	Exact address and documents are never in public documents or public URLs	TC-MAP-008, TC-DOC-009
NFR-REL-01	Reliability	A failed notification never fails the main action	TC-BOOK-018
NFR-REL-02	Reliability	Back up before every migration or rules change	TC-OPS-003
NFR-PERF-01	Performance	Browse first contentful paint < 2.5 s on 4G mid-range phone; JS route chunks lazy-loaded	Lighthouse mobile ≥ 80
NFR-PERF-02	Performance	Confirm API p95 < 2 s (warm), < 5 s (cold start)	API logs
NFR-ACC-01	Accessibility	WCAG 2.2 AA on user pages	axe: 0 serious issues
NFR-QUOTA-01	Capacity	A full demo day stays under 50% of Firestore Spark daily quotas	Usage tab after rehearsal

ID	Category	Requirement	Measure
NFR-MAINT-01	Maintainability	Lint, unit tests and build pass on every PR to dev	CI green

10. Honest limitations (use these exact words)

- "Document review is a **plausibility check**. We match names and look for obvious tampering. It is not a legal title search with the Registry of Deeds."
- "We **do not hold deposits**. Money moves directly between renter and owner. We provide the accountability trail and the consequences."
- "The bank catalog is a **curated public scrape** of Metrobank listings, with a visible 'Last verified' date. It is not a bank partnership."
- "Notifications are **in-app only** in this version."
- "Demo data is **synthetic** and labelled as such. No real IDs or titles are stored."

11. Glossary

Term	Meaning
Verified – document reviewed	Badge for a listing whose documents passed admin plausibility review
Trusted API	The small serverless backend (Vercel) that performs actions clients must not perform themselves
listingPrivate	Firestore collection holding the exact address and document references, readable by owner and admins only
Geohash (5-char)	Encoded map cell of about 4.9 km × 4.9 km, used for the public approximate pin
ROPA	Real and Other Properties Acquired: foreclosed bank assets
Spark / Blaze	Firebase's free plan / pay-as-you-go plan (Blaze requires a billing account)
F-xx	Defect ID from the live test findings (e.g., F-05 booking overlap, F-16 document access)